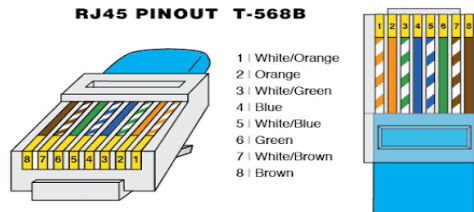


Network cable and connectors

Task – 1 Cat-6 Ethernet cable color code

- Pin 1 :- White
- Pin 2 :- Orange
- Pin 3 :- Green
- Pin 4 :- Blue
- Pin 5 :- White
- Pin 6 :- Green
- Pin 7 :- Brown
- Pin 8 :- Dark Brown



Task – 2 Different types of ethernet cables

- Cat5 :- Retail and restaurant point-of-sale systems require low bandwidth for simple data transactions (like sending orders or billing receipts). Cat5e easily supports speeds up to 1 Gbps at a very low cost, making it ideal and economical for everyday indoor equipment.
- Cat6 :- Modern Wi-Fi routers handle high-density traffic and support high bandwidth demands across dozens of connected devices. Cat6 features a central plastic spline that reduces signal crosstalk, allowing it to deliver reliable 10 Gbps speeds over shorter distances (up to 55 meters).
- Fiber cable :- Live video streaming platforms require massive, ultra-fast data transmission with near-zero latency to broadcast high-definition video to millions of concurrent viewers. Fiber optic cables use light pulses instead of electrical signals, allowing them to carry huge amounts of data over vast distances without electromagnetic interference.

Task – 3 How to crimp an RJ-45 connector onto a network cable.

1. Strip the Outer Cover :- Use a cable stripper or the cutting blade on your crimper to gently remove about an inch (2.5 to 4 cm) of the outer plastic jacket from the cable end. Be careful not to slice into the thin insulation on the small wires inside.
2. Untwist and Straighten the Wires :- Separate the four pairs of twisted wires and untwist them completely. Use your fingers to straighten every single wire so they lie flat and smooth next to each other. If you are using a Cat6 cable, snip off the plastic divider sticking out of the middle.
3. Line Up and Trim the Wires :- Arrange the wires in order from left to right using the standard T568B color pattern. Hold them tightly between your fingers so they stay in place. Then, cut straight across all the wires, leaving about half an inch (1.2 cm) of exposed wire sticking out from the plastic jacket.
4. Slide the Wires into the Connector :- Hold the RJ-45 plug with the clip facing down (gold pins facing up). Carefully push the straightened wires straight into the plug. Make sure all eight wires push all the way to the very front tip and that the outer plastic jacket goes slightly inside the back of the plug.
5. Crimp the Connector :- Put the RJ-45 plug into the crimping tool's slot. Squeeze the handles together firmly until the tool clicks or locks down completely. This pushes the gold contacts into the wires and locks the plastic plug onto the cable.
 - Mistake to avoid :- Don't in the different wiring in the different slot. And, do perfect wiring bcoz it can make noise and problem in slot.

Task – 4 Choosing the perfect wiring to connect on ipl event.

- Display Connections (Consoles to Screens) :- HDMI 2.1 Cables: Essential for current-gen consoles (PS5, Xbox Series X) to output 4K at 120Hz or 8K for live IPL video streams.
- Fiber Optic HDMI Cables :-Used for runs longer than 5 meters (e.g., from central consoles to elevated projector screens or main stage displays). Unlike standard copper, active optical cables maintain full 48 Gbps bandwidth over long distances without signal degradation.
- Network Connections (Multiplayer & Live Streaming) :- Cat6 or Cat6a Ethernet Cables: Handles local console networking, tournament play, and high-bitrate live video streams without Wi-Fi congestion or jitter.
- RJ-45 Connectors :- The standard termination for Ethernet cables, ensuring a secure, click-locked physical link to consoles, routers, and switches.

- Audio Output (Event Sound System) :- Routes low-latency multi-channel audio from screens or audio extractors directly to the venue's PA system or soundbars.

Task – 5 Connector or network cable

- Cable & Connector Type :- Cable (Cat6) Twisted-Pair Ethernet Cable.
Connector: Standard 8-position, 8-contact RJ-45 clear plastic connector with gold-plated contact pins.